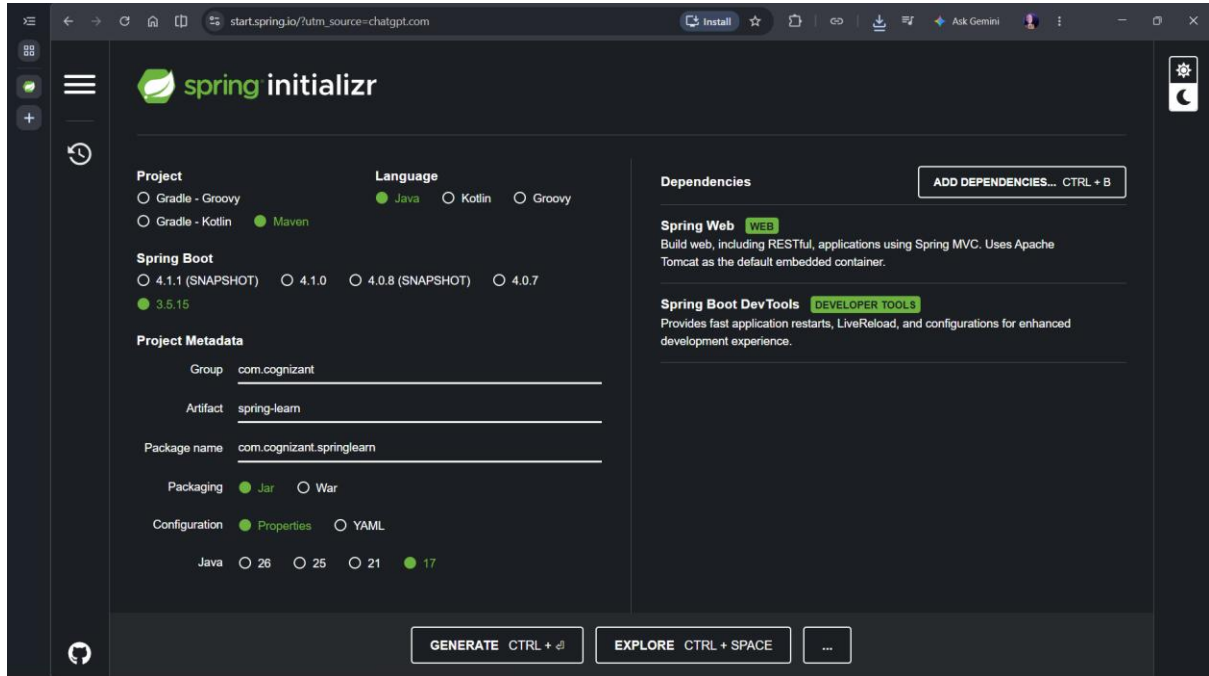
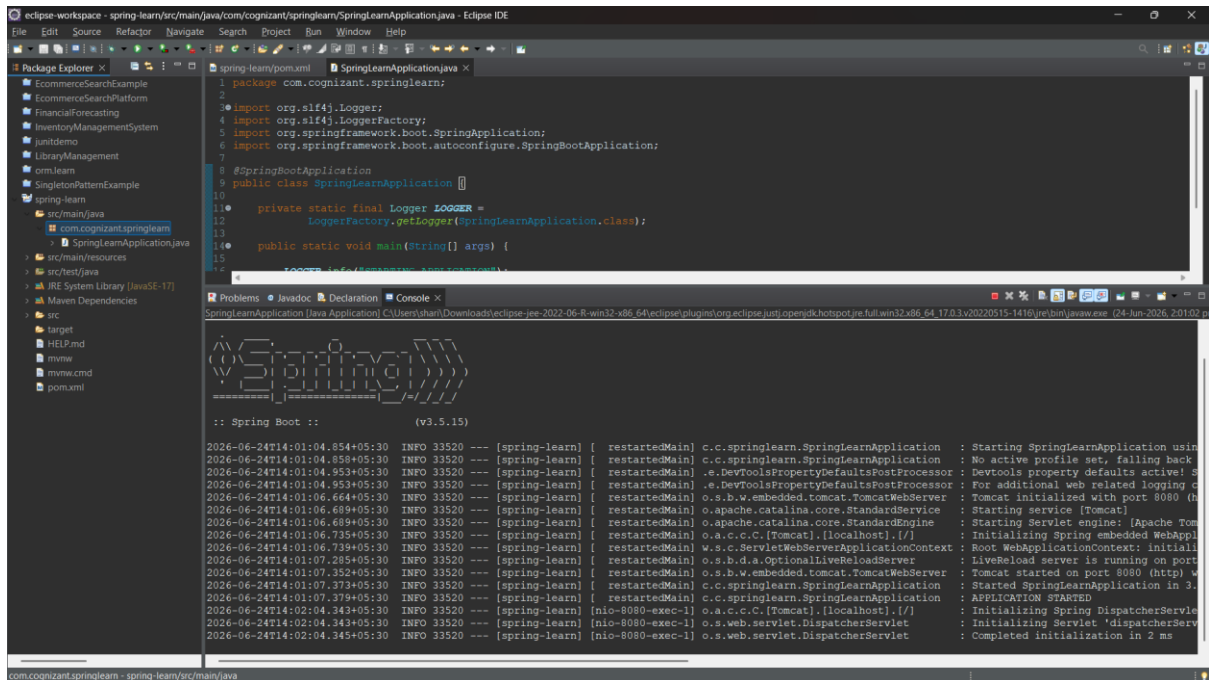


Create a Spring Web Project using Maven

Creating spring project



OUTPUT



Spring Core – Load SimpleDateFormat from Spring Configuration XML

date-format.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"

       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="

           http://www.springframework.org/schema/beans

           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="dateFormat"

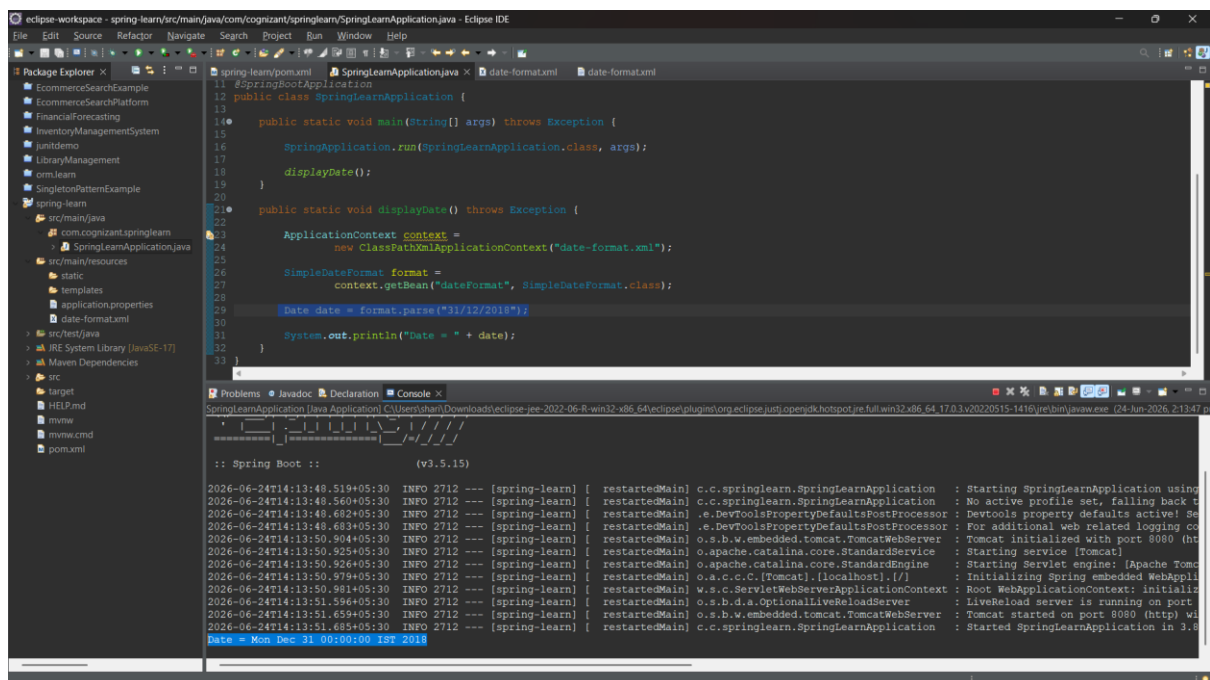
          class="java.text.SimpleDateFormat">

        <constructor-arg value="dd/MM/yyyy"/>

    </bean>

</beans>
```

OUTPUT



```
Spring Boot :: (v3.5.15)

2026-06-24T14:13:48.519+05:30 INFO 2712 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using
2026-06-24T14:13:48.560+05:30 INFO 2712 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to default profile: 'default'
2026-06-24T14:13:48.682+05:30 INFO 2712 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults active! See https://aka.ms/spring-devtools for additional configuration details.
2026-06-24T14:13:48.683+05:30 INFO 2712 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging configuration, see https://aka.ms/spring-devtools#web.
2026-06-24T14:13:50.904+05:30 INFO 2712 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-06-24T14:13:50.926+05:30 INFO 2712 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.90]
2026-06-24T14:13:50.979+05:30 INFO 2712 --- [spring-learn] [ restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-06-24T14:13:50.981+05:30 INFO 2712 --- [spring-learn] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed
2026-06-24T14:13:51.596+05:30 INFO 2712 --- [spring-learn] [ restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 6060
2026-06-24T14:13:51.659+05:30 INFO 2712 --- [spring-learn] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path ''
2026-06-24T14:13:51.685+05:30 INFO 2712 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 3.84 seconds (JVM running for 4.15s)

Date = Mon Dec 31 00:00:00 IST 2018
```

Hello World RESTful Web Service

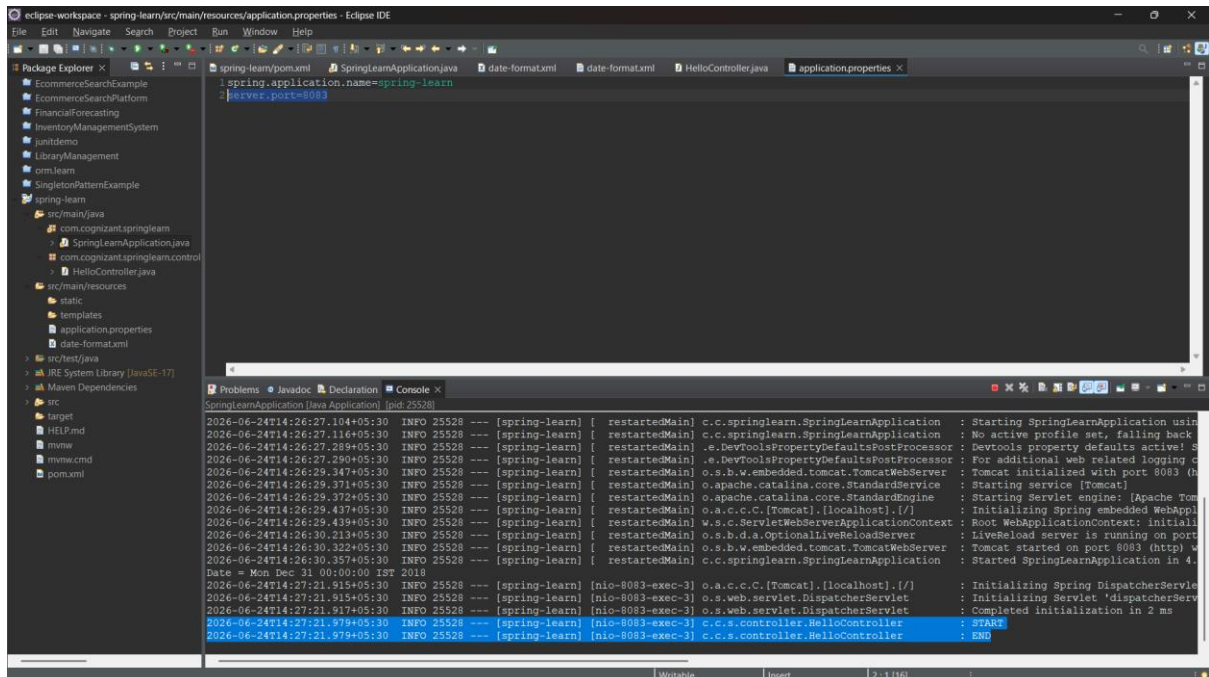
HelloController.java

```
package com.cognizant.springlearn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
```

@RestController

```
public class HelloController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(HelloController.class);
    @GetMapping("/hello")
    public String sayHello() {
        LOGGER.info("START");
        String message = "Hello World!!";
        LOGGER.info("END");
        return message;
    }
}
```

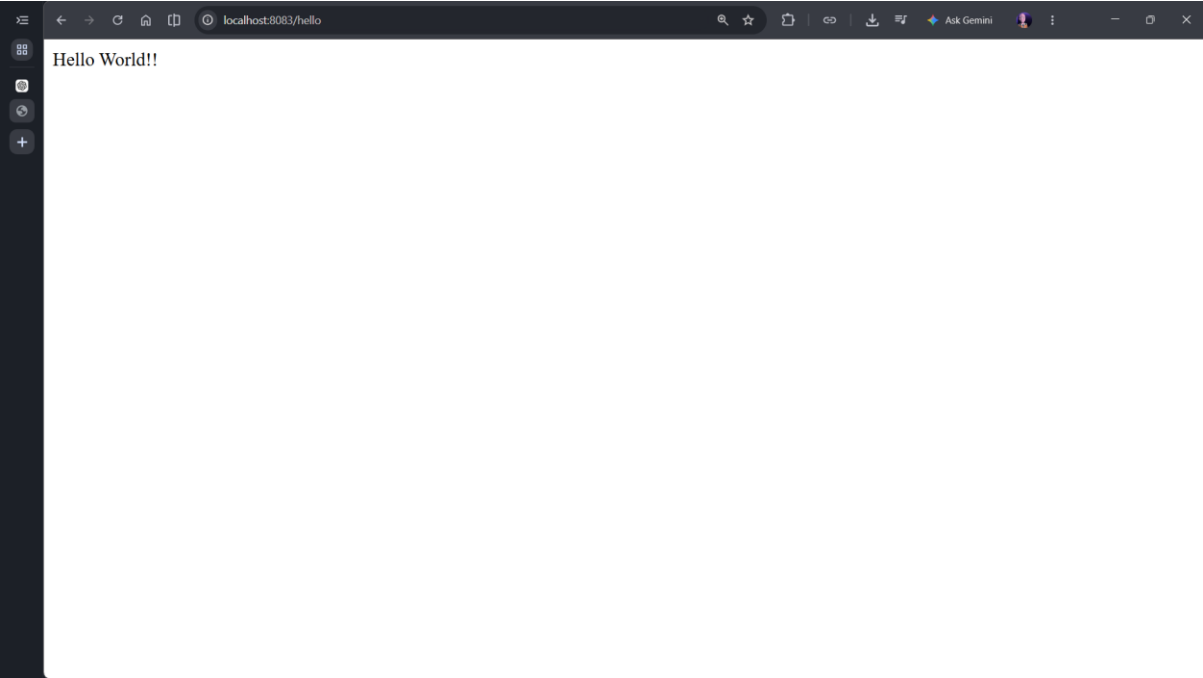
OUTPUT



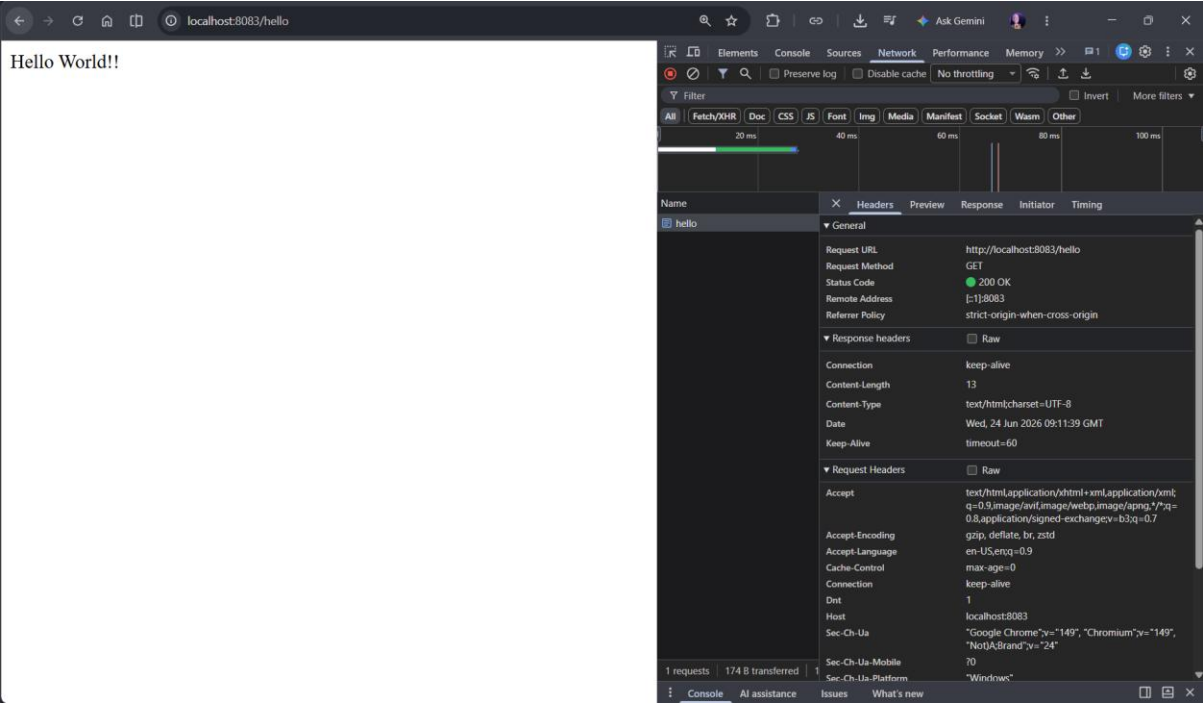
The screenshot shows the Eclipse IDE interface. The Package Explorer on the left displays the project structure, including the 'spring-learn' project and its sub-packages. The main editor area shows the 'SpringLearnApplication.java' file. The Console window at the bottom displays the output of the application, showing the startup sequence and the successful execution of the 'sayHello' endpoint.

```
SpringLearnApplication [Java Application] [pid: 25528]
2026-06-24T14:26:27.104+05:30 INFO 25528 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Starting SpringLearnApplication using 'dev' profile
2026-06-24T14:26:27.116+05:30 INFO 25528 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : No active profile set, falling back to default profiles: dev
2026-06-24T14:26:27.289+05:30 INFO 25528 --- [spring-learn] [restartedMain] e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults are active. For additional web related logging check the 'logging.level.org.springframework.devtools.web' property
2026-06-24T14:26:27.290+05:30 INFO 25528 --- [spring-learn] [restartedMain] e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging check the 'logging.level.org.springframework.devtools.web' property
2026-06-24T14:26:29.347+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-06-24T14:26:29.371+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Starting service [Tomcat]
2026-06-24T14:26:29.372+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.56]
2026-06-24T14:26:29.437+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-06-24T14:26:29.439+05:30 INFO 25528 --- [spring-learn] [restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization started
2026-06-24T14:26:30.213+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port 35728
2026-06-24T14:26:30.322+05:30 INFO 25528 --- [spring-learn] [restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path ''
2026-06-24T14:26:30.357+05:30 INFO 25528 --- [spring-learn] [restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 4.188 seconds
Date = Mon Dec 31 00:00:00 IST 2018
2026-06-24T14:27:21.915+05:30 INFO 25528 --- [spring-learn] [nio-8083-exec-3] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet
2026-06-24T14:27:21.915+05:30 INFO 25528 --- [spring-learn] [nio-8083-exec-3] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-06-24T14:27:21.917+05:30 INFO 25528 --- [spring-learn] [nio-8083-exec-3] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
2026-06-24T14:27:21.919+05:30 INFO 25528 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.HelloController : START
2026-06-24T14:27:21.979+05:30 INFO 25528 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.HelloController : END
```

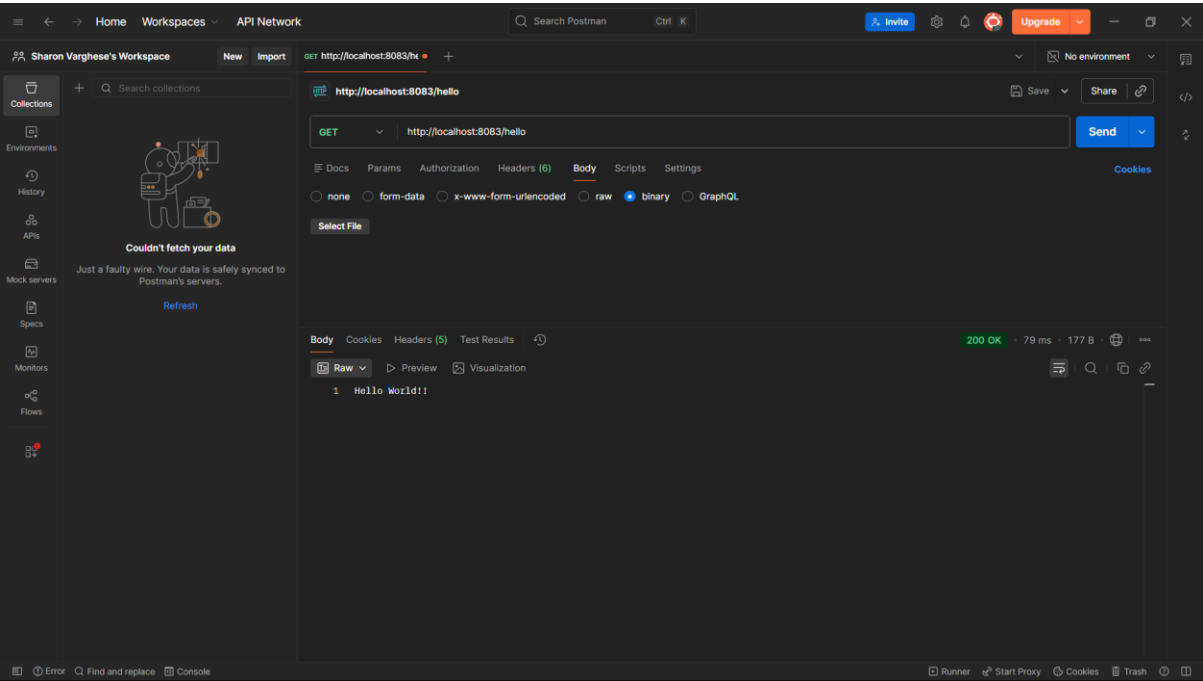
WEB OUTPUT



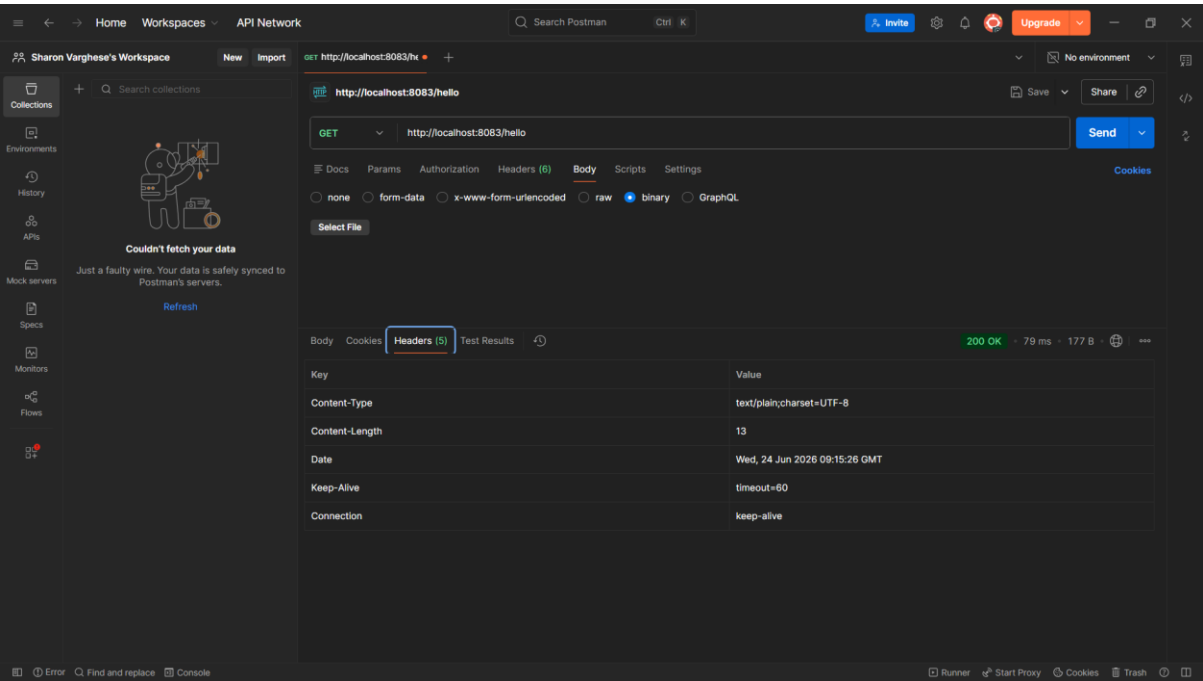
DEVELOPER TOOLS OUTPUT



POSTMAN OUTPUT



POSTMAN HEADER

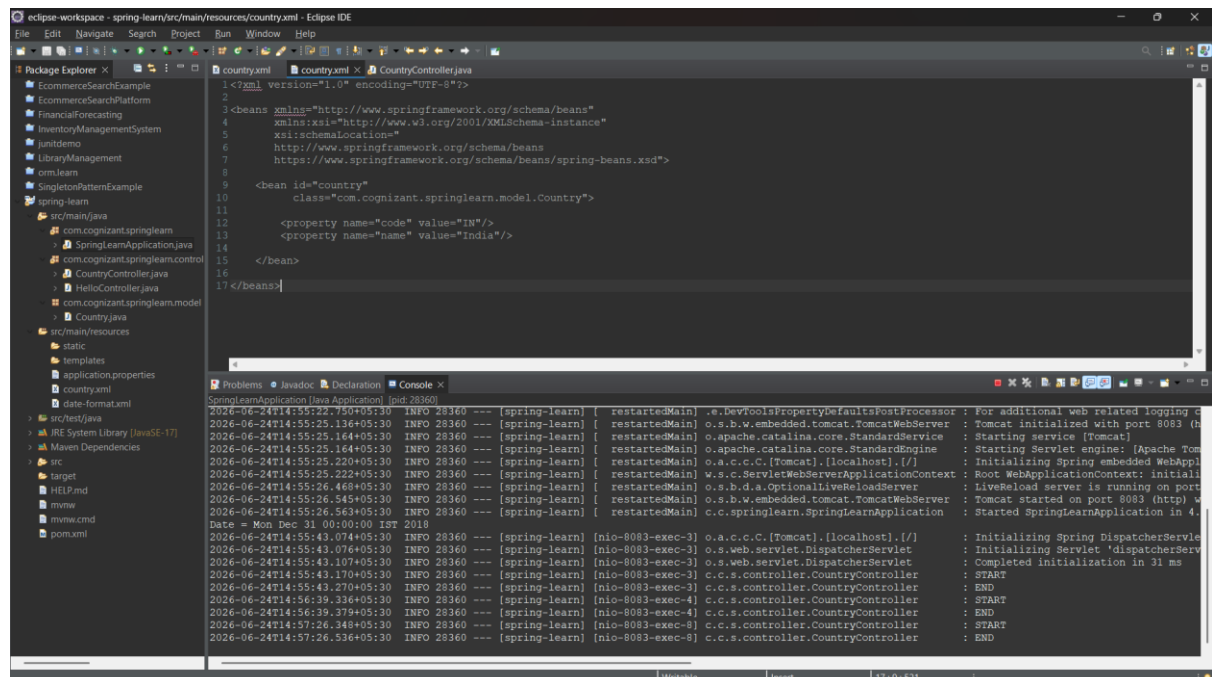


REST - Country Web Service

CountryController.java

```
package com.cognizant.springlearn.controller;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.model.Country;
@RestController
public class CountryController {
    private static final Logger LOGGER =
        LoggerFactory.getLogger(CountryController.class);
    @RequestMapping("/country")
    public Country getCountryIndia() {
        LOGGER.info("START");
        ApplicationContext context =
            new ClassPathXmlApplicationContext("country.xml");
        Country country =
            context.getBean("country", Country.class);
        LOGGER.info("END");
        return country;
    }
}
```

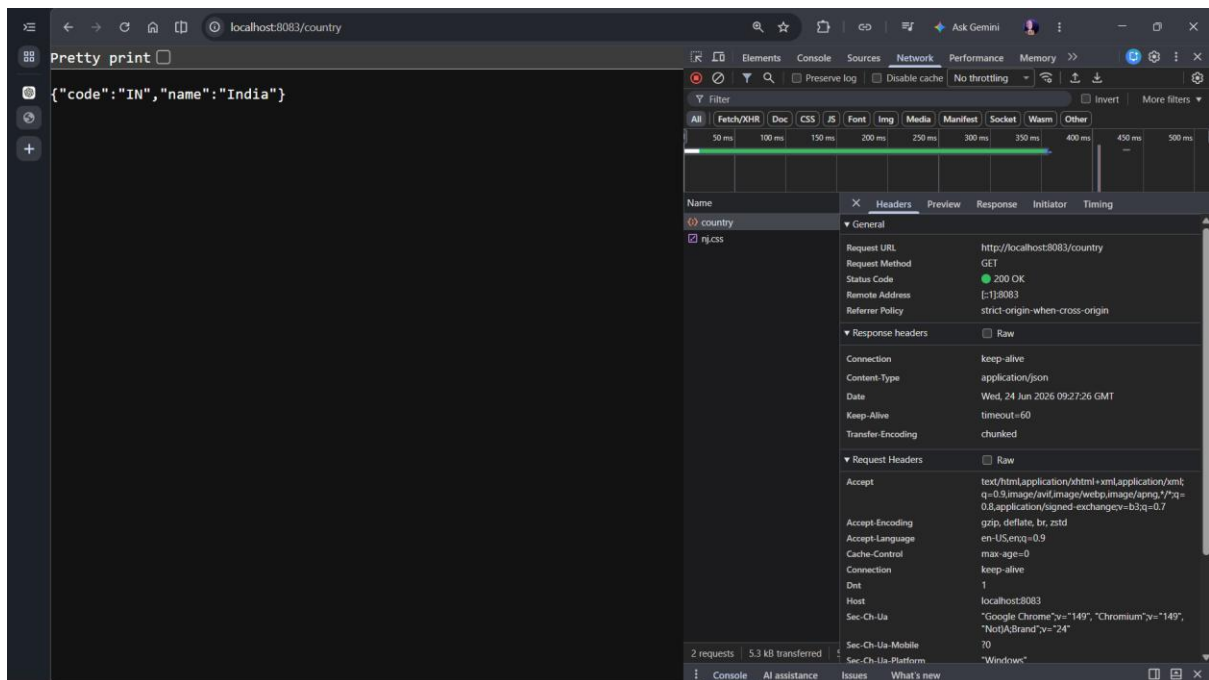
OUTPUT



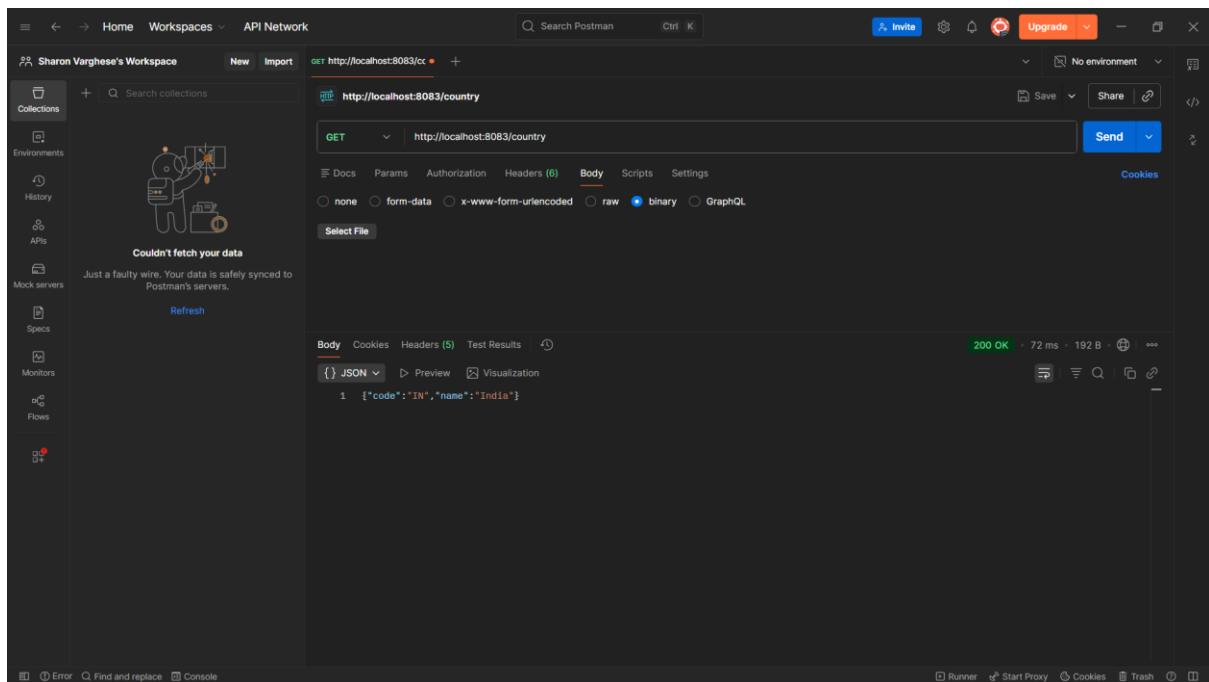
The screenshot displays the Eclipse IDE interface. The left sidebar shows the Package Explorer with the project structure. The main editor window shows the CountryController.java file. The bottom console window displays the output of the application, including the start and end of the getCountryIndia() method, and the initialization of the Spring application.

```
2026-06-24T14:55:22.750+05:30 INFO 28360 --- [spring-learn] restartedMain] .e.befoolsPropertyDefaultsPostProcessor : For additional web related logging c
2026-06-24T14:55:25.136+05:30 INFO 28360 --- [spring-learn] restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (h
2026-06-24T14:55:25.164+05:30 INFO 28360 --- [spring-learn] restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-24T14:55:25.164+05:30 INFO 28360 --- [spring-learn] restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tom
2026-06-24T14:55:25.220+05:30 INFO 28360 --- [spring-learn] restartedMain] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebAppli
2026-06-24T14:55:25.222+05:30 INFO 28360 --- [spring-learn] restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initiali
2026-06-24T14:55:26.468+05:30 INFO 28360 --- [spring-learn] restartedMain] o.s.b.d.a.OptionalLiveReloadServer : LiveReload server is running on port
2026-06-24T14:55:26.545+05:30 INFO 28360 --- [spring-learn] restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8083 (http://0.0.0.0:8083)
2026-06-24T14:55:26.563+05:30 INFO 28360 --- [spring-learn] restartedMain] c.c.springlearn.SpringLearnApplication : Started SpringLearnApplication in 4.
Date = Mon Dec 31 00:00:00 IST 2018
2026-06-24T14:55:43.074+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-3] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServic
2026-06-24T14:55:43.076+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-3] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServ
2026-06-24T14:55:43.107+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-3] o.s.web.servlet.DispatcherServlet : Completed initialization in 31 ms
2026-06-24T14:55:43.170+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.CountryController : START
2026-06-24T14:55:43.270+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-3] c.c.s.controller.CountryController : END
2026-06-24T14:56:39.336+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-4] c.c.s.controller.CountryController : START
2026-06-24T14:56:39.379+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-4] c.c.s.controller.CountryController : END
2026-06-24T14:57:26.348+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-8] c.c.s.controller.CountryController : START
2026-06-24T14:57:26.536+05:30 INFO 28360 --- [spring-learn] [nio-8083-exec-8] c.c.s.controller.CountryController : END
```

WEB HEADER

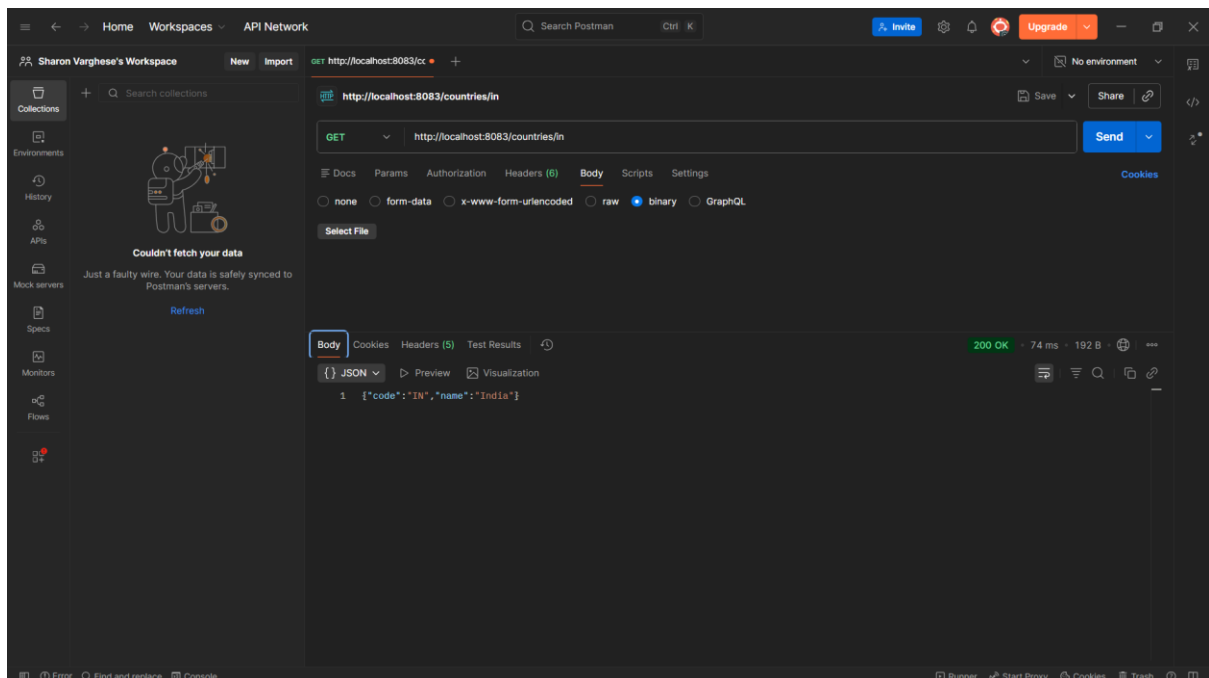
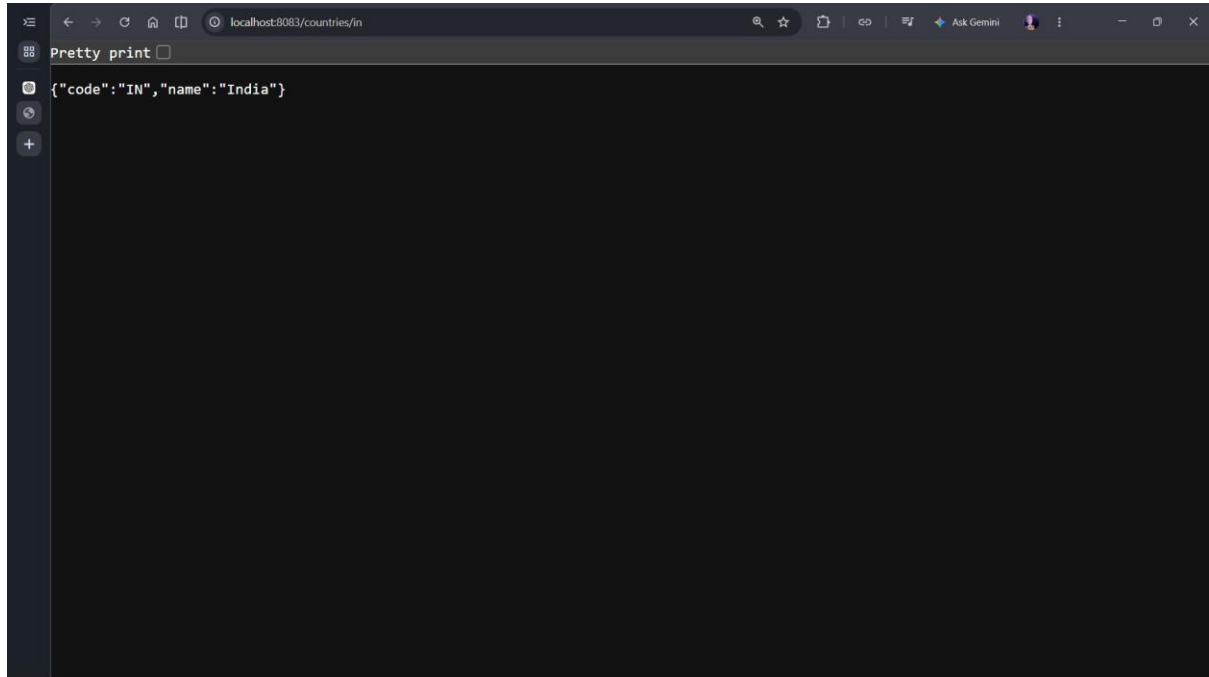


POSTMAN OUTPUT



REST - Get country based on country code

OUTPUT



Create authentication service that returns JWT

AuthenticationController.java

```
package com.cognizant.springlearn.controller;
import java.util.Base64;
import java.util.Date;
import java.util.HashMap;
import java.util.Map;
import javax.crypto.SecretKey;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestHeader;
import org.springframework.web.bind.annotation.RestController;
import io.jsonwebtoken.Jwts;
import io.jsonwebtoken.SignatureAlgorithm;
import io.jsonwebtoken.security.Keys;
@RestController
public class AuthenticationController {
    @GetMapping("/authenticate")
    public Map<String, String> authenticate(
        @RequestHeader("Authorization") String authHeader) {
        Map<String, String> map = new HashMap<>();
        String user = getUser(authHeader);
        String token = generateJwt(user);
        map.put("token", token);
        return map;
    }
    private String getUser(String authHeader) {
        String encodedCredentials =
            authHeader.substring("Basic ".length());
        byte[] decodedBytes =
            Base64.getDecoder().decode(encodedCredentials);
        String credentials =
            new String(decodedBytes);
        return credentials.split(":")[0];
    }
    private String generateJwt(String user) {
        SecretKey key =
            Keys.hmacShaKeyFor(
                "secretkeysecretkeysecretkey12345"
                    .getBytes());
        return Jwts.builder()
            .setSubject(user)
            .setIssuedAt(new Date())
            .setExpiration(
                new Date(System.currentTimeMillis()
                    + 1200000))
            .signWith(key, SignatureAlgorithm.HS256)
            .compact();
    }
}
```

CMD REQUEST OUTPUT

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shari>curl -s -u user:pwd http://localhost:8083/authenticate
{"token":"eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJlc2VyaWwif0IjoxNzgyMjk4ODAsLCJleHAiOiJlZ300IzMDAwMDI9.IlJYh00IEupu1kFW9FP-6871288WEyeyo5EENXKy-u0"}
C:\Users\shari>
```